

**Project Title:- AI-Powered Real-Time
Interview Analysis and Evaluation System
(InterViewAI Coach)**

(By SEMESTER – VIII of IV Year M.Sc. (CA & IT) -2026)

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Table of Contents

1. ABSTRACT.....	3
2. OBJECTIVE OF PROJECT	4
3. TOOLS / TECHNOLOGY USED.....	5
4. METHODOLOGY / ALGORITHM	6
5. SYSTEM FLOWCHART / DESIGN	8
6. IMPEMENTATION	10
• Screenshots	10
• Important Code	13
7. RESULT / OUTPUT.....	16
8. CONCLUSION.....	18
9. REFERENCES	19
10. APPENDIX	20
• GitHub Link:-.....	20

1. ABSTRACT

The AI-Based Interview Analysis System is developed to automate and improve the interview evaluation process using Artificial Intelligence and Soft Computing techniques. Traditional interviews often depend on human judgment, which may introduce bias and inconsistency. Our system solves this issue by evaluating candidates based on their communication skills, confidence, fluency, and technical responses.

The system analyzes candidate performance using three major factors: answer analysis, audio analysis, and video analysis. Audio analysis evaluates speech confidence, fluency, clarity, fillers, pauses, and speaking speed using signal processing techniques. Video analysis evaluates eye contact, facial visibility, smile ratio, head movement, and stability to estimate confidence level during the interview.

The project uses Firestore database for storing user, interview, and question data. Cloudinary is used for storing interview videos, while Google Drive is used for storing generated reports. The system also supports multiple difficulty levels (Easy, Medium, Hard), where higher levels are unlocked based on performance.

The proposed system provides automatic scoring, detailed feedback, confidence evaluation, and downloadable reports, making interview assessment more efficient, objective, and intelligent.

2. OBJECTIVE OF PROJECT

The main objective of this project is to develop an intelligent interview evaluation system that can analyze candidate performance automatically using Artificial Intelligence and Soft Computing techniques.

Objectives:

- **To create a realistic interview environment** Provide users with an experience similar to real-life interviews.
- **To analyze user performance in real-time** Evaluate responses based on speech, confidence, and timing.
- **To improve communication skills** Help users enhance clarity, confidence, and speaking ability.
- **To generate automated feedback** Provide instant results without the need for human evaluators.
- **To record and store interview sessions** Maintain data for review and continuous improvement.
- **To make interview preparation more accessible** Allow users to practice anytime and from anywhere.

3. TOOLS / TECHNOLOGY USED

Technology / Tool	Purpose
React.js	Frontend development and user interface
Python	Backend processing and AI analysis
Firebase Firestore	Database management and data storage
Librosa	Audio analysis and feature extraction
OpenCV	Video frame processing
MediaPipe	Face landmark detection
Sentence Transformers	Semantic answer evaluation
FFmpeg	Audio extraction from video
Cloudinary	Video storage

4. METHODOLOGY / ALGORITHM

The system follows a multi-stage methodology for interview analysis:

Step 1: User Identification

The system checks whether the user already exists in Firestore database. Existing user details are fetched, otherwise a new user profile is created.

Step 2: Level Selection

The candidate selects the interview level:

- Easy (10 minutes)
- Medium (15 minutes)
- Hard (20 minutes)

The next level gets unlocked only when the user scores above 80%.

Step 3: Question Fetching

Questions are fetched dynamically from Firestore based on selected domain and difficulty level.

Step 4: Live Interview Processing

During the interview:

- Camera remains active
- Audio recording takes place
- Questions are displayed one by one
- Candidate can answer or skip questions

Step 5: Audio Analysis

Audio is analyzed using Librosa to extract:

- Energy
- Pitch
- Fluency
- Speaking speed
- Fillers
- Pauses

Step 6: Video Analysis

Video frames are processed every 3rd frame using MediaPipe and OpenCV for:

- Eye contact
- Smile detection
- Face visibility
- Head movement
- Stability

Step 7: Answer Analysis

Candidate answers are evaluated using:

1. Keyword Matching
2. Semantic Similarity
3. Code Structure Matching (for coding questions)

Step 8: Final Score Generation

➤ Confidence Formula:

$$\text{Confidence} = 0.6(\text{Audio}) + 0.4(\text{Video})$$

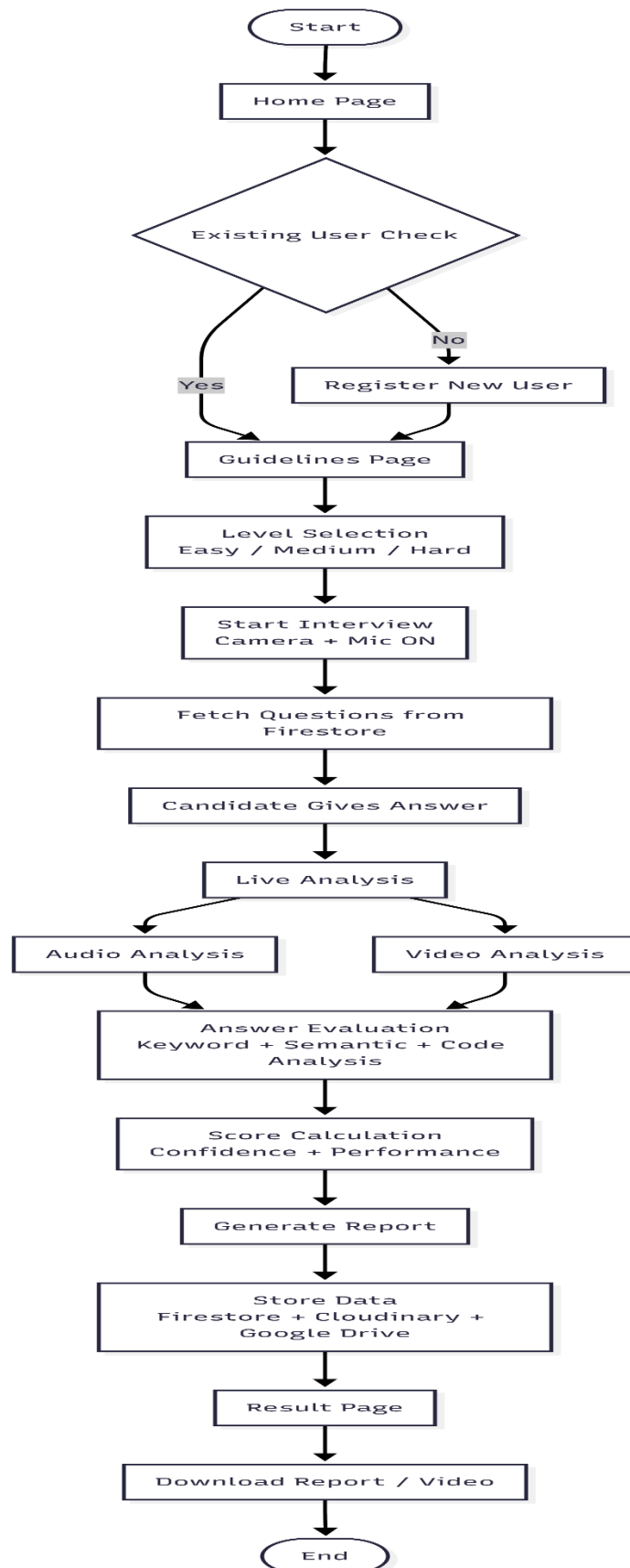
➤ Performance Formula:

$$\text{Performance} = 0.4(\text{Answer}) + 0.3(\text{Audio}) + 0.3(\text{Video})$$

Step 9: Report Generation

The final report is generated and stored in Google Drive, while interview videos are uploaded to Cloundinary.

5. SYSTEM FLOWCHART / DESIGN



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The AI-Based Interview Analysis System follows a structured workflow to automate interview evaluation using answer analysis, audio analysis, and video analysis. The system processes user information, conducts interviews, performs real-time analysis, and generates a final performance report.

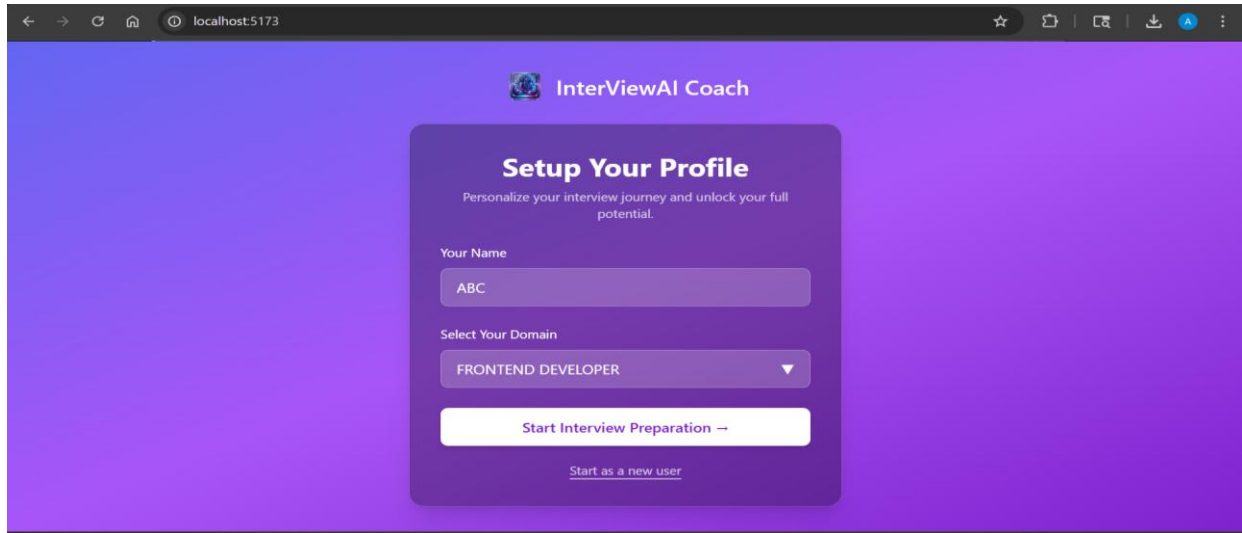
The system is divided into multiple modules including frontend interface, backend processing, database management, AI-based analysis, and cloud storage.

6. IMPEMENTATION

Screenshots

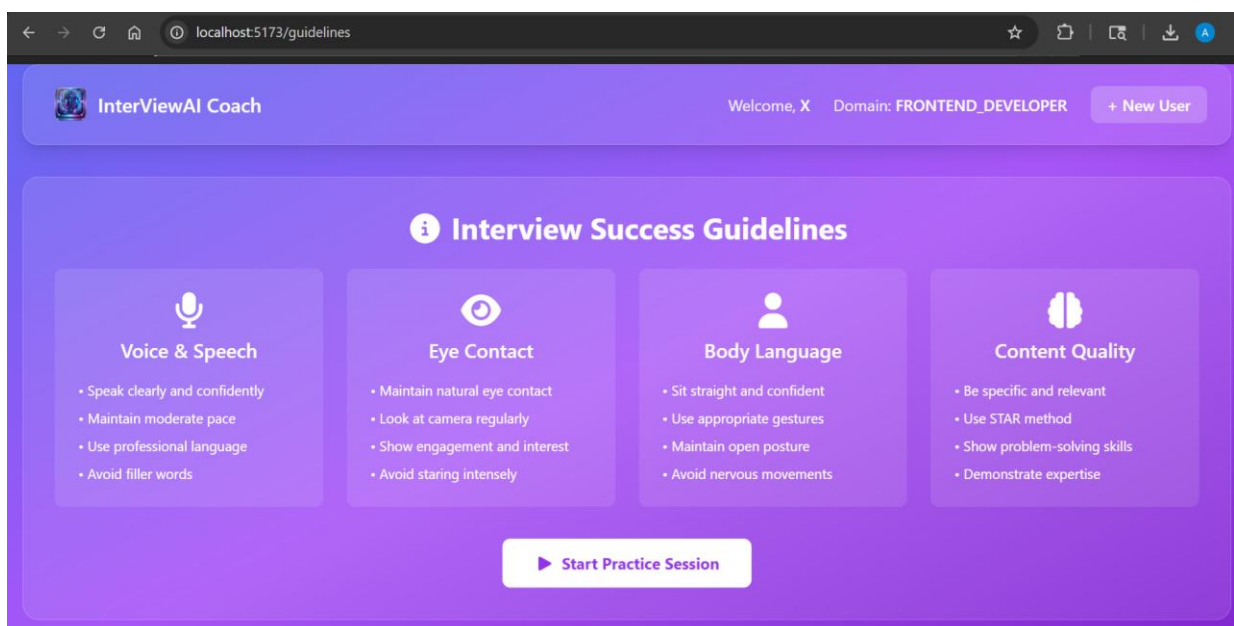
➤ Home Page

The home page verifies whether the user already exists in the database. Existing user details such as name and domain are fetched from Firestore. If the user is new, a new profile is created.



➤ Guidelines Page

This page displays interview instructions and guidelines for candidates before starting the interview process.

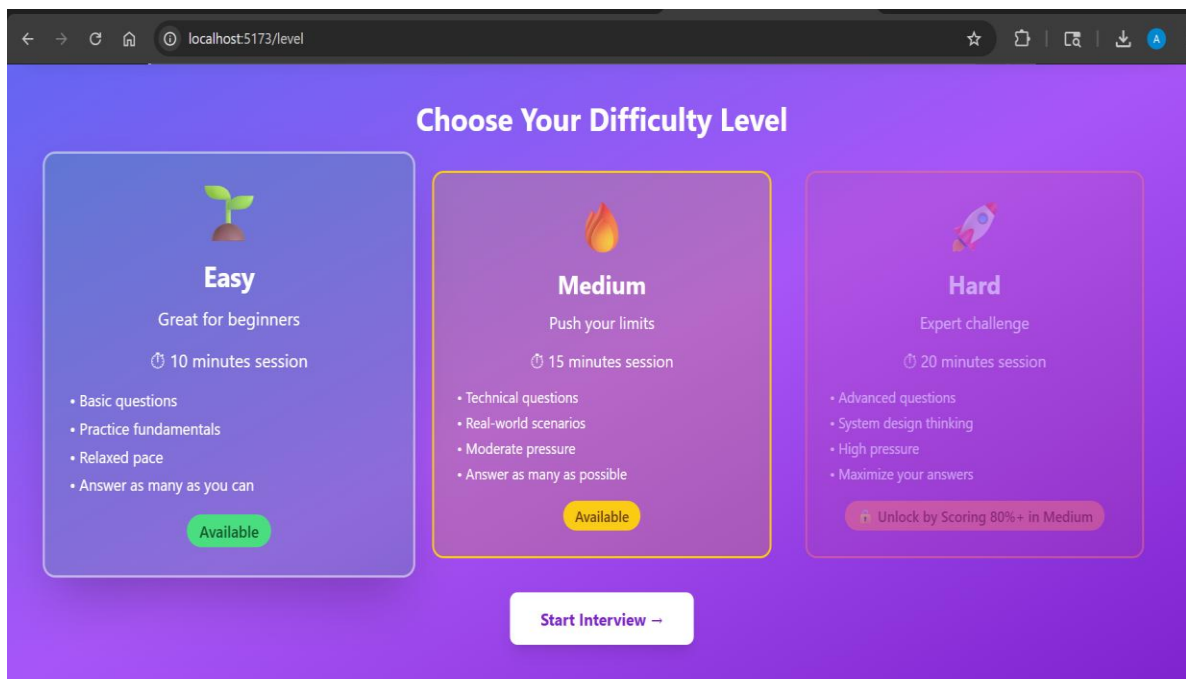


➤ Level Selection Page

The system provides three interview levels:

- Easy (10 Minutes)
- Medium (15 Minutes)
- Hard (20 Minutes)

The next level unlocks only if the candidate scores above 80%.

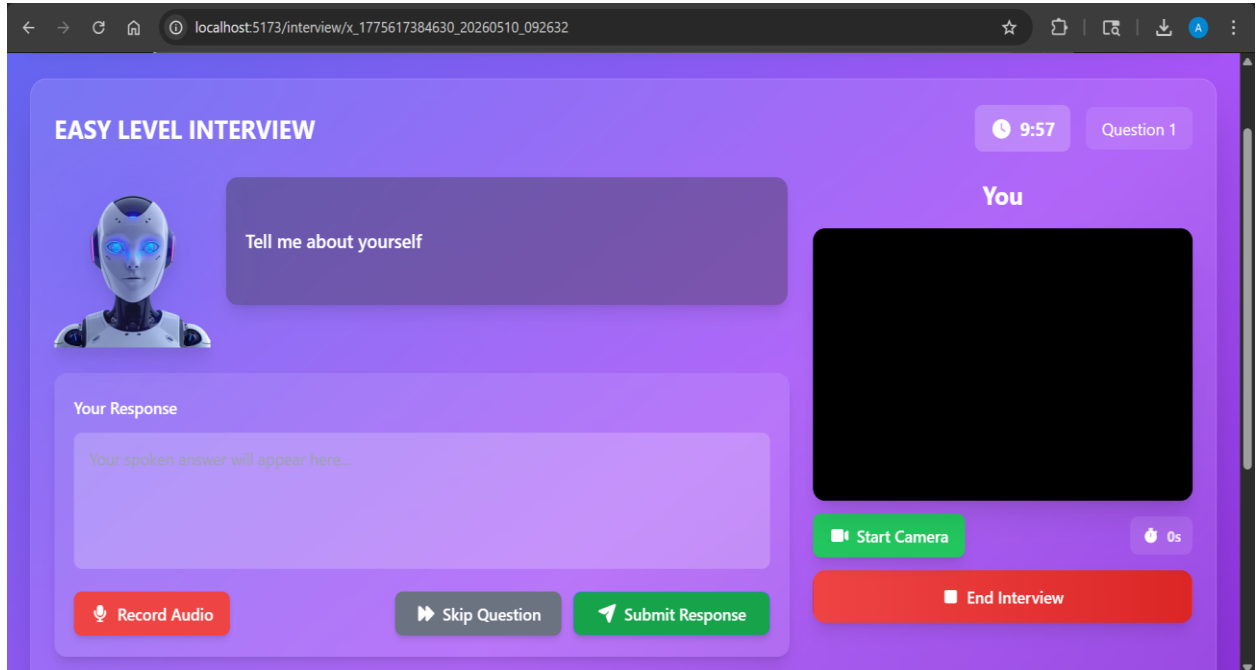


➤ Interview Page

During the interview:

- Camera and microphone are activated
- Questions are fetched dynamically from Firestore
- Candidate answers questions
- Skip option available
- Live audio and video analysis performed

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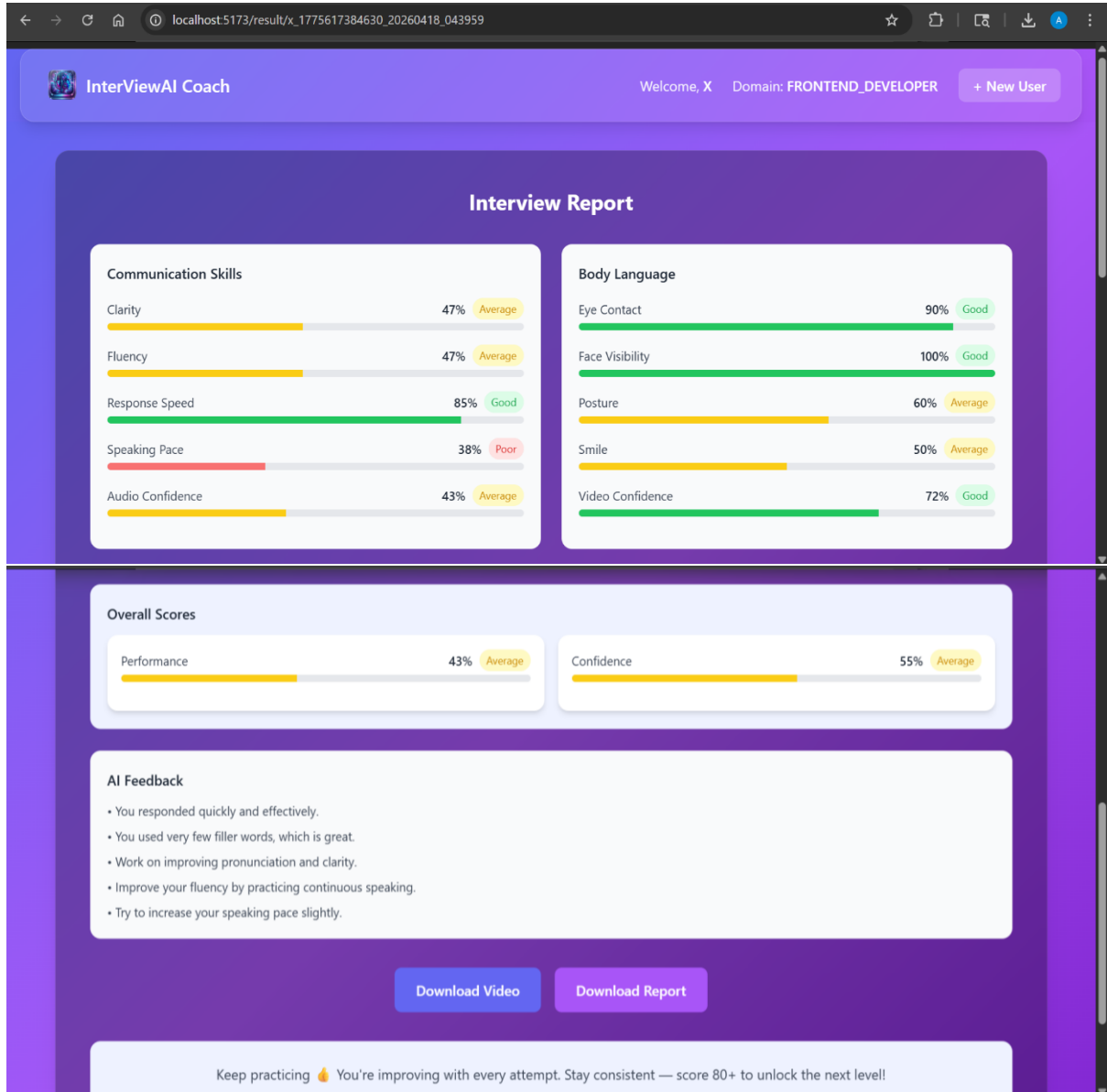


➤ **Result Page**

After interview completion, the result page displays:

- Audio analysis
- Video analysis
- Confidence score
- Performance score
- Feedback
- Download report/video option

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Important Code

1. Answer Evaluation Logic

The system evaluates answers using keyword matching and semantic similarity.

$$\text{final} = (0.6 * \text{semantic_score}) + (0.4 * \text{keyword_score})$$

2. Audio Analysis Score Calculation

Audio confidence is calculated using multiple speech features.

```
overall = (  
    (energy_score * 0.20) +  
    (fluency_score * 0.40) +  
    (clarity_score * 0.20) +  
    (speed_score * 0.20)  
)
```

Explanation:

- Energy → confidence
- Fluency → speaking smoothness
- Clarity → speech quality
- Speed → speaking pace

3. Video Analysis Score

Video confidence is calculated using facial behavior.

```
video_confidence = (  
    eye_score * 0.4 +  
    expression_score * 0.25 +  
    stability_score * 0.2 +  
    visibility * 10 * 0.15  
)
```

Explanation:

- Eye contact → confidence
- Smile → expression analysis
- Stability → movement analysis
- Visibility → face detection

4. Final Confidence Score

```
confidence_score = (audio_score * 0.6) +  
(video_score * 0.4)
```

5. Performance Score

```
performance_score = (  
    answer_score * 0.4 +  
    audio_score * 0.3 +  
    video_score * 0.3  
)
```

7. RESULT / OUTPUT

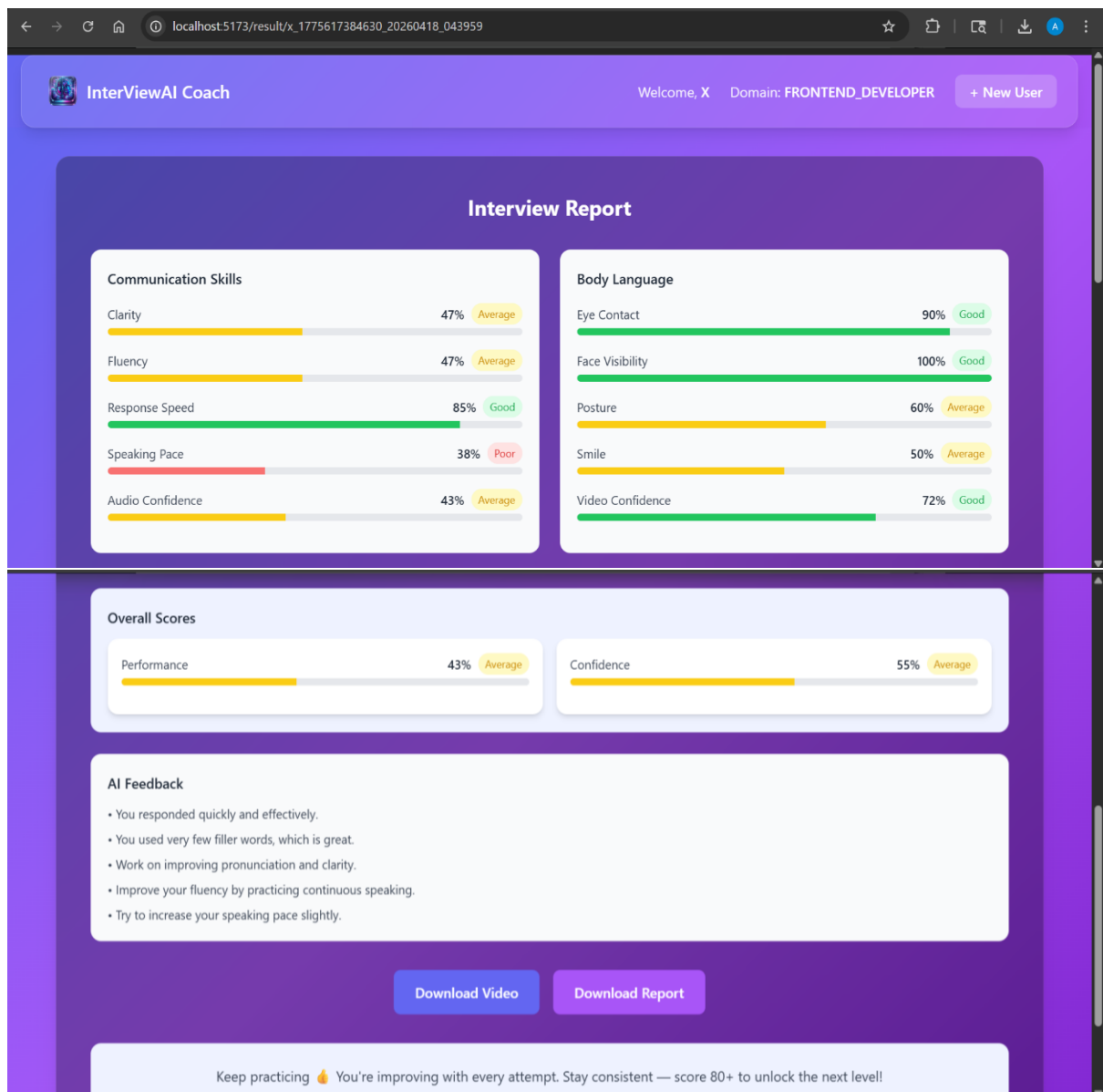
The AI-Based Interview Analysis System successfully evaluates candidate performance by analyzing answers, speech, and facial behavior during the interview process. The system generates confidence and performance scores based on multiple parameters and provides detailed feedback to the candidate.

Output Generated by the System

The system provides the following outputs:

- **Answer Analysis**
 - Keyword matching
 - Semantic similarity evaluation
 - Coding answer analysis
- **Audio Analysis**
 - Clarity
 - Fluency
 - Speaking speed
 - Fillers detection
 - Audio confidence score
- **Video Analysis**
 - Eye contact detection
 - Smile detection
 - Face visibility
 - Head movement analysis
 - Stability score
- **Final Performance Evaluation**
 - Confidence Score
 - Performance Score
 - Overall Feedback
- **Additional Features**
 - Interview report generation
 - Downloadable video
 - Downloadable report

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8. CONCLUSION

The AI-Based Interview Analysis System successfully automates the interview evaluation process using Artificial Intelligence and Soft Computing techniques. The system evaluates candidates based on answer quality, communication skills, and behavioral confidence through audio and video analysis.

By combining semantic answer evaluation, facial analysis, and speech processing, the system reduces manual effort and human bias in interviews. The generated reports and confidence scores provide a structured and objective assessment of candidate performance.

The project demonstrates how AI can improve interview evaluation through intelligent and real-time analysis.

9. REFERENCES

- Python Documentation – <https://docs.python.org/3/>
- React.js Documentation – <https://react.dev/>
- Librosa Documentation – <https://librosa.org/doc/latest/index.html>
- OpenCV Documentation – <https://docs.opencv.org/>
- MediaPipe Documentation – <https://developers.google.com/mediapipe>
- Firebase Firestore Documentation – <https://firebase.google.com/docs/firestore>

10. APPENDIX

 **GitHub Link:-**

<https://github.com/pooja877/Interviewer-.git>